

535/2  
PHYSICS  
Paper 2  
(Practical)  
May / June 2026  
2 Hours



ASK INTEGRATED TEACHERS' EXAMINATIONS BUREAU LTD

Uganda Lower Secondary Certificate of Education

S.4 PRE MOCK 1 ASSESSMENTS 2026

PHYSICS

PAPER 2

**INSTRUCTIONS TO CANDIDATES:**

*This paper consists of **two examination items**.*

*Respond to **one item only**.*

*Any additional item responded to will not be scored.*

*You are not allowed to start working with the apparatus for the first **15 minutes**.  
This time is to enable you to read the items thoroughly, check whether you have all  
the apparatus needed for the investigation, and plan appropriately.*

*Responses to the item must be written in the **answer booklet(s) provided**.*

*Indicate the **item number** you are responding to clearly at the top of the page.*

***Graph paper** is provided.*

***Mathematical tables** and **silent non-programmable calculators** may be used.*

## Item 1

A local **solar energy company** is facing challenges with the efficiency of its solar concentrators. Engineers suspect that the lenses being used are not of the correct refractive index, which reduces the ability of the concentrators to focus sunlight effectively.

The company has acquired a glass block from which new lenses are to be made, but they want to confirm that the material has a refractive index suitable for solar applications (between 1.40 and 1.59).

### Task

You are provided with a glass block of the same optical properties as the one the company received. Carry out a scientific investigation to ascertain its appropriateness for use in solar concentrators.

(Hint: In your report, include the aim, variables, hypothesis, procedure, sources of error, precautions, data presentation with at least 5 sets, data analysis using a graph, and your recommendation.)

## Item 2

A **new brand of motorcycles** has been introduced in Uganda. The transport safety authority requires that the suspension springs of these motorcycles must have a force constant between **20 Nm<sup>-1</sup> and 40 Nm<sup>-1</sup>** to ensure rider safety and stability on rough roads.

During inspection, it is observed that some springs appear too soft or too stiff, raising concerns about whether they meet the required specifications.

### Task

You are provided with a spring made of the same material as the motorcycle suspension springs. Plan and carry out an investigation to determine whether it meets the safety requirements.

(Hint: In your report, include the aim, variables, hypothesis, procedure, sources of error, precautions, data presentation with at least 5 sets, data analysis using a graph, and your recommendation.)

**END**

## CONFIDENTIAL

### Item 1

- ✓ Glass block
- ✓ Optical pins (**4 pieces**)
- ✓ Plain paper
- ✓ Soft board
- ✓ Thumb pins (**2 pieces**)
- ✓ Metre rule
- ✓ Protractor
- ✓ Pencil
- ✓ Graph paper

### Item 2

- Spring
- Mass hanger (50 g capacity)
- Set of masses ( $5 \times 50$  g)
- Retort stand with clamp
- Metre rule
- Thread (50 cm)
- Stopwatch
- Graph paper